

设 $f(x,y)$ 在 $D \subseteq \mathbb{R}^2$ 上非负, 对于两类广义二重积分. 若按某一确定方式取 D_1 , 使 $D_1 \rightarrow D$ 时 $\iint_{D_1} f(x,y) dx dy$ 以 A 为极限, 则 $\iint_D f(x,y) dx dy$ 收敛于 A

计算 $\iint_D e^{-(x^2+y^2)} dx dy$, $D = \{(x,y) | x \geq 0, y \geq 0\}$

找区域 D_1

取 $D_1 = \{(x,y) | x^2+y^2 \leq r^2, x \geq 0, y \geq 0\}$

$$\iint_D e^{-(x^2+y^2)} dx dy = \lim_{D_1 \rightarrow D} \iint_{D_1} e^{-(x^2+y^2)} dx dy$$

$$= \lim_{r \rightarrow +\infty} \int_0^{\frac{\pi}{2}} d\theta \int_0^r e^{-\rho^2} \rho d\rho = \lim_{r \rightarrow +\infty} \frac{\pi}{4} (1 - e^{-r^2}) = \frac{\pi}{4}$$

计算 $\int_0^{+\infty} e^{-x^2} dx$

$$I^2 = \int_0^{+\infty} e^{-x^2} dx \cdot \int_0^{+\infty} e^{-y^2} dy = \int_0^{+\infty} dx \int_0^{+\infty} e^{-(x^2+y^2)} dy = \frac{\pi}{4}$$

$$\Rightarrow I = \frac{\sqrt{\pi}}{2}$$

计算 $\iint_D \frac{1}{\sqrt{x^2+y^2}} dx dy$, 其中 $D = \{(x,y) | x^2+y^2 \leq 1\}$

$$\iint_D \frac{1}{\sqrt{x^2+y^2}} dx dy = \lim_{\varepsilon \rightarrow 0^+} \iint_{D_\varepsilon} \frac{1}{\sqrt{x^2+y^2}} dx dy = \lim_{\varepsilon \rightarrow 0^+} \int_\varepsilon^1 dr \int_0^{2\pi} \frac{1}{\sqrt{r^2}} r d\theta = 2\pi(1-\varepsilon) \Rightarrow \pi$$

计算 $\int_0^1 \frac{x^b - x^a}{\ln x} dx \quad (b > a \geq 0)$

$$\begin{aligned} \int_0^1 \frac{x^b - x^a}{\ln x} dx &= \int_0^1 dx \int_a^b x^y dy = \int_a^b dy \cdot \int_0^1 x^y dx \\ &= \int_a^b \left. \frac{x^{y+1}}{y+1} \right|_{x=0}^{x=1} dy = \ln \frac{b+1}{a+1} \end{aligned}$$

计算 $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx$ 换元 + 区间再现法

令 $x = \tan \theta$

$$I = \int_0^{\frac{\pi}{4}} \ln(1 + \tan \theta) d\theta$$

令 $\theta = \frac{\pi}{4} - \varphi$

$$I = \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1 + \tan \varphi}\right) d\varphi$$

$$2I = \int_0^{\frac{\pi}{4}} \ln 2 d\theta = \frac{\pi}{4} \ln 2$$

$$\Rightarrow I = \frac{\pi}{8} \ln 2$$

计算 $\int_0^{+\infty} \frac{\sin x}{x} dx$

$$\frac{1}{x} = \int_0^{+\infty} e^{-tx} dt$$

$$I = \int_0^{+\infty} \sin x \left(\int_0^{+\infty} e^{-tx} dt \right) dx$$

$$\begin{aligned} \text{收敛} &= \int_0^{+\infty} \left(\int_0^{+\infty} e^{-tx} \sin x dx \right) dt \\ &= \int_0^{+\infty} \frac{1}{1+t^2} dt = \frac{\pi}{2} \end{aligned}$$